Daily Logic Drill #5 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Spot the Flaw · Common Fallacies · Proof Critique · Valid vs. Invalid Arguments.

Section A

1. A student solves $\sqrt{x-2} = -3$ by squaring both sides to obtain $x - 2 = 9 \implies x = 11$. Does $x = 11$ satisfy the original equation?

Answer: False.

Method: Substituting $x = 11$ gives $\sqrt{11-2} = \sqrt{9} = 3 \neq -3$. Squaring both sides introduces an extraneous solution where the two sides originally had opposite signs. Thus $x = 11$ does not satisfy the equation.

> **Investigate further:** Under what condition on the right-hand side does $\sqrt{f(x)} = c$ have any real solutions?

2. A student simplifies $\dfrac{x^2-4}{x-2} = 4 \implies x + 2 = 4 \implies x = 2$. Is $x = 2$ a valid solution to the original equation?

Answer: False.

Method: At $x = 2$, the denominator $x-2$ equals 0, so the expression $\frac{x^2-4}{x-2}$ is undefined. The cancellation $\frac{x^2-4}{x-2} = x + 2$ is valid only for $x \neq 2$. Thus $x = 2$ is not in the domain of the equation, leaving it with no solutions.

> **Investigate further:** What type of discontinuity does the function $g(x) = \frac{x^2-4}{x-2}$ have at $x = 2$?

3. In an attempted proof that $1 = 2$, a student starts with $a = b$, obtains $(a - b)(a + b) = b(a - b)$, and concludes $a + b = b$. Which algebraic operation is invalid?

Answer: Dividing by $a - b$ (division by zero is invalid).

Method: Since $a = b$, the factor $a - b$ is equal to 0. Dividing both sides of $(a - b)(a + b) = b(a - b)$ by $a - b$ constitutes division by zero, which is undefined.

> **Investigate further:** Why does dividing by zero allow one to prove any arbitrary equality such as $1 = 2$?

4. A student claims that for all real x, $x^2 > 4 \implies x > 2$. Give a negative integer counterexample to this deduction.

Answer: -3

Method: For $x = -3$, $x^2 = 9 > 4$ holds, but $-3 > 2$ is false. Any $x \leq -2$ disproves the claim.

> **Investigate further:** What is the correct equivalence for $x^2 > 4$?

5. A student solves $\log(x)+\log(x-3) = 1$ and finds candidate solutions $x = -2$ and $x = 5$. Which candidate is extraneous?

Answer: -2

Method: For $x = -2$, neither $\log(-2)$ nor $\log(-5)$ is defined for real logarithms, which require strictly positive arguments. Only $x = 5$ is valid: $\log(5) + \log(2) = \log(10) = 1$.

> **Investigate further:** Why does combining $\log(a)+\log(b) = \log(ab)$ expand the domain of definition?

6. A student conjectures: "If $f'(c) = 0$, then f has a local maximum or minimum at $x = c$." Give a polynomial function $f(x)$ with $c = 0$ that disproves this claim.

Answer: $f(x) = x^3$ (an inflection point, no local extremum).

Method: For $f(x) = x^3$, $f'(x) = 3x^2 \implies f'(0) = 0$. However, $x = 0$ is a stationary point of inflection: $f(x) < 0$ for $x < 0$ and $f(x) > 0$ for $x > 0$, so f has neither a local maximum nor a local minimum at $x = 0$.

> **Investigate further:** What condition on higher derivatives determines whether a stationary point is a local extremum?

7. A student asserts: "$|a| < |b| \implies a < b$." Provide a counterexample pair of integers (a, b).

Answer: $(1, -2)$

Method: For $a = 1$ and $b = -2$, we have $|a| = 1 < 2 = |b|$, but $1 < -2$ is false.

> **Investigate further:** When does $|a| < |b| \implies a < b$ hold?

8. A student concludes that because n^2 is divisible by 4, n must be divisible by 4. Find the smallest positive integer counterexample n.

Answer: 2

Method: For $n = 2$, $n^2 = 4$, which is divisible by 4, but 2 is not divisible by 4. The smallest counterexample is $n = 2$.

> **Investigate further:** If p is prime and $p \mid n^2$, must $p \mid n$?

9. A student claims: "For all real numbers a, b, if $a < b$ then $a^2 < b^2$." Give a pair of integers (a, b) with $a < b$ and $a^2 > b^2$.

Answer: (-5, 2)

Method: For $a = -5$ and $b = 2$, $-5 < 2$ holds, but $a^2 = 25 > 4 = b^2$.

> **Investigate further:** Under what condition on a and b does $a < b \implies a^2 < b^2$ hold?

10. A student asserts: "If $A \subseteq B \cup C$, then $A \subseteq B$ or $A \subseteq C$." Give a set A of two elements and sets $B = \{1\}, C = \{2\}$ that disprove this claim.

Answer: $A = \{1, 2\}$

Method: Let $A = \{1, 2\}$. Then $A \subseteq \{1\} \cup \{2\} = \{1, 2\}$ holds, but $A \not\subseteq \{1\}$ and $A \not\subseteq \{2\}$.

> **Investigate further:** Is it true that if $x \in B \cup C$, then $x \in B$ or $x \in C$?

Section B

1. A student attempts to prove: "For all integers n, if n is odd, then n^2 is odd."
Line 1: Suppose n is odd, so $n = 2k + 1$ for some integer k.
Line 2: Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.
Line 3: Since $2k^2 + 2k$ is an integer, n^2 is odd.
Is this proof valid? Justify your answer.

Answer: Yes, fully valid.

Method: Every line follows correctly: the algebra in Line 2 is correct ($(2k+1)^2 = 4k^2 + 4k + 1$), and Line 3 correctly identifies the result as $2 \times (\text{integer}) + 1$, the definition of odd. There is no flaw.

> **Investigate further:** Is this a direct proof or a proof by contrapositive of some related statement?

2. Find the flaw: "Claim: If $x^2 = 3x$, then $x = 3$." Proof: Divide both sides of $x^2 = 3x$ by x to get $x = 3$.

Answer: Dividing by x is invalid when $x = 0$.

Method: $x^2 = 3x \iff x(x-3) = 0 \iff x = 0$ or $x = 3$. Dividing by x silently assumes $x \neq 0$ and loses the solution $x = 0$, which also satisfies the original equation.

> **Investigate further:** Rewrite the claim correctly, including both solutions.

3. Find the flaw: "Claim: The only real solution to $\sqrt{x+7} = x - 5$ is $x = 9$." Proof: Squaring both sides gives $x + 7 = x^2 - 10x + 25$, so $x^2 - 11x + 18 = 0$, so $(x-9)(x-2) = 0$, giving $x = 9$ or $x = 2$.

Answer: $x = 2$ is an extraneous root introduced by squaring; it must be checked against and rejected from the original equation.

Method: Checking $x = 2$ in the original equation: $\sqrt{2+7} = \sqrt{9} = 3$, but $x - 5 = 2 - 5 = -3$. Since $3 \neq -3$, $x = 2$ does not satisfy the original equation. Squaring introduced this extraneous solution; only $x = 9$ (which does check: $\sqrt{16} = 4 = 9 - 5$) is valid.

> **Investigate further:** Why does squaring specifically introduce solutions where the two sides had opposite sign?

4. A student argues: For every integer n, $6\left(\frac{4n+1}{2} - \frac{n-3}{2}\right)$ is a multiple of 3. *(after TMUA 2020 Paper 2 Q3)*
Line 1: $6\left(\frac{4n+1}{2} - \frac{n-3}{2}\right) = 3\left(2\left(\frac{4n+1}{2} - \frac{n-3}{2}\right)\right)$.
Line 2: $= 3(4n + 1 - n - 3)$.
Line 3: $= 3(3n - 2)$.
Line 4: which is always a multiple of 3.
On which line does the argument first contain an error, and what exactly is that error?

Answer: Line 2

Method: Correctly distributing, $2\left(\frac{4n+1}{2} - \frac{n-3}{2}\right) = (4n+1) - (n-3) = 4n + 1 - n + 3 = 3n + 4$, not

$4n + 1 - n - 3$ as written. The sign of -3 was not flipped when distributing the subtraction across the bracket $(n - 3)$. (Interestingly, since Line 1's outer factor of 3 is untouched by this error, the final conclusion — a multiple of 3 — remains true regardless; the argument is still invalid even though its conclusion happens to be correct.)

> **Investigate further:** Compute the genuinely correct simplified form, $3(3n + 4)$, and confirm it is still a multiple of 3.

5. Find the flaw: "Claim: For every integer n, $n^2 + n + 2$ is even." Proof: If n is even, $n = 2k$, so $n^2 + n + 2 = 4k^2 + 2k + 2 = 2(2k^2 + k + 1)$, which is even. Hence $n^2 + n + 2$ is even for every integer n.

Answer: The proof only checks the even case; the odd case is never addressed.

Method: To prove a claim for "every integer n", both parities must be checked. (The claim does happen to also hold for odd n: $n = 2k+1 \implies n^2+n+2 = 4k^2+4k+1+2k+1+2 = 4k^2+6k+4 = 2(2k^2+3k+2)$, even — but the proof as given never establishes this.)

> **Investigate further:** Complete the missing odd case explicitly.

6. Find the flaw: "Claim: For real x, $x^2 = 9 \iff x = 3$." Proof: If $x = 3$, then $x^2 = 9$. This proves the biconditional.

Answer: Only the $\Longleftarrow$ direction ($x = 3 \implies x^2 = 9$) was shown; the $\Longrightarrow$ direction ($x^2 = 9 \implies x = 3$) is false, since $x = -3$ also satisfies $x^2 = 9$.

Method: A biconditional requires both directions to be proved. The correct statement is $x^2 = 9 \iff x = 3$ or $x = -3$.

> **Investigate further:** What is the correct biconditional relating $x^2 = 9$ to a single equation in x?

7. Find the flaw: "Claim: For every positive integer n, the n-th Fibonacci number F_n (with $F_1 = F_2 = 1$) is either equal to 1 or prime." Proof: $F_3 = 2$, $F_4 = 3$, $F_5 = 5$ — all prime. Since the pattern holds for $n = 3, 4, 5$, the claim is proved. What is the smallest counterexample?

Answer: Checking finitely many cases does not prove a universal claim; $F_6 = 8 = 2^3$ is the smallest counterexample.

Method: $F_1 = 1, F_2 = 1, F_3 = 2, F_4 = 3, F_5 = 5, F_6 = 8$. $F_6 = 8$ is neither equal to 1 nor prime $(8 = 2 \times 4)$, disproving the claim. Three verified cases give no logical guarantee about the fourth.

> **Investigate further:** Is F_n prime for infinitely many n? This remains an open problem in general.

8. A student wants to prove: "If n is not divisible by 3, then n^2 is not divisible by 3." Instead, they show: "If n is divisible by 3, then n^2 is divisible by 3." Explain why this does not prove the original claim.

Answer: The statement shown is neither the original claim nor its contrapositive — it proves nothing about the original.

Method: Original claim: $\neg A \implies \neg B$ where A: "$3 \mid n$", B: "$3 \mid n^2$". Its valid contrapositive is $B \implies A$ ("if $3 \mid n^2$ then $3 \mid n$"). The student instead proved $A \implies B$, a different (though also true) statement, which establishes neither the original claim nor its contrapositive.

> **Investigate further:** State the actual contrapositive of the original claim, and outline how you would prove it.

9. Find the flaw: "Claim: For all integers a, b, if a and b are both even, then $a + b$ is even." Proof: Since $a + b$ is even, $a + b = 2m$ for some integer m. Since a, b are even, $a = 2j, b = 2k$, so $2j + 2k = 2m$, i.e. $j + k = m$, which is true since m is an integer. Hence $a + b$ is even.

Answer: Circular reasoning: the proof begins by assuming $a + b$ is even (the very thing to be proved) instead of deriving it.

Method: A correct proof starts from $a = 2j, b = 2k$ and derives $a + b = 2j + 2k = 2(j + k)$, which is even because $j + k$ is an integer — it never needs to assume $a + b = 2m$ up front.

> **Investigate further:** Rewrite this as a correct, non-circular direct proof.

10. A student attempts to solve $\log_2(x - 1) + \log_2(x + 1) = 3$:
Line 1: $\log_2((x - 1)(x + 1)) = 3$.
Line 2: $(x - 1)(x + 1) = 2^3 = 8$.
Line 3: $x^2 - 1 = 8 \implies x^2 = 9$.
Line 4: $x = 3$ or $x = -3$.
Line 5: Hence the solution set is $\{-3, 3\}$.
On which line does the first error occur, and what is the true solution set? *(after TMUA 2017 Paper 2 Q13)*

Answer: Line 5 (only $x = 3$ is valid in the domain).

Method: Lines 1–4 correctly deduce $x = \pm 3$. However, in the original equation $\log_2(x-1)+\log_2(x+1) = 3$, the terms require $x - 1 > 0$ and $x + 1 > 0$, so $x > 1$. The candidate $x = -3$ makes the arguments negative ($\log_2(-4)$ and $\log_2(-2)$), which are undefined. Line 5 fails to verify the domain restriction; the true solution set is $\{3\}$.

> **Investigate further:** Why does combining $\log(a) + \log(b) = \log(ab)$ expand the domain to include points where both a and b are negative?

Section C

1. Consider the claim: "For every positive integer n, $2n^2 + 2n + 1$ is prime." *(after TMUA 2022 Paper 2 Q7)*

Answer: D

Method: Lines I–III are correct: the discriminant of $2x^2 + 2x + 1$ is indeed -4, so the polynomial has no real (let alone integer linear) factorisation. But Line IV commits a category error: a polynomial being irreducible over $\mathbb{Z}[x]$ says nothing about whether its integer OUTPUT VALUES are prime. At $n = 3$: $2(9) + 6 + 1 = 25 = 5^2$, composite, disproving the original claim.

> **Investigate further:** Find another value of n for which $2n^2 + 2n + 1$ is composite.

2. Let P: "$x^2 - 5x + 6 = 0$" and Q: "$x = 2$". A student argues $Q \implies P$ is true and concludes $P \iff Q$.

Answer: B

Method: $x^2 - 5x + 6 = (x - 2)(x - 3) = 0$, so P is satisfied by $x = 2$ or $x = 3$. $Q \implies P$ is indeed true, but $P \implies Q$ is false since $x = 3$ satisfies P without satisfying Q. Asserting $P \implies Q$ merely because the statements "seem related" is the flaw.

> **Investigate further:** What is the correct relationship (necessary/sufficient) between P and Q?

3. A student attempts to prove: "For every integer n, $n^2 + n$ is divisible by 2", checking only the even case.

Answer: B

Method: $n^2 + n = n(n+1)$ is a product of two consecutive integers, so it is even regardless of the parity of n — but the given proof only demonstrates this for even n and never addresses odd n, so as written it is incomplete.

> **Investigate further:** Complete the missing odd case: for $n = 2k + 1$, show $n^2 + n$ is even.

4. A student attempts to prove: "No three consecutive odd positive integers can all be prime." *(after TMUA 2023 Paper 2 Q4)*

Answer: C

Method: Lines I and II are correct: among $n-2, n, n+2$, exactly one is always divisible by 3. Line III is the flaw — being a positive multiple of 3 only implies compositeness if the multiple is *greater than* 3; the multiple could equal 3 itself, which is prime. Exactly this happens at $n = 5$: the three consecutive odd integers $3, 5, 7$ are all prime.

> **Investigate further:** Are there any other counterexamples to the original claim besides $3, 5, 7$?

5. A student "proves" $\forall n\, \exists m\, (m > n)$ and concludes $\exists M\, \forall n\, (M > n)$.

Answer: B

Method: Swapping the order of a universal and existential quantifier is not a valid logical step in general. $\forall n\, \exists m\, (m > n)$ says every n has SOME larger m (true, e.g. $m = n + 1$); $\exists M\, \forall n\, (M > n)$ says a SINGLE M exceeds every n simultaneously, which is false (no integer is larger than every integer).

> **Investigate further:** Give a general example of a true $\forall\exists$ statement whose $\exists\forall$ swap is false.

6. A student "confirms" the converse of "rhombus $\Longrightarrow$ perpendicular diagonals" using only the example of a square.

Answer: B

Method: One example (the square) supports the converse but cannot prove a universal statement. A kite has perpendicular diagonals but is generally not a rhombus, disproving the converse.

> **Investigate further:** Sketch a kite that is not a rhombus, and confirm its diagonals are perpendicular.

7. A student disproves $(*)$: "For every pair of real numbers x, y, $|x + y| < |x| + |y|$" using the pair $x = -2, y = -5$. *(after TMUA 2020 Paper 2 Q8)*

Answer: D

Method: This is a trap: the student's reasoning is actually completely correct. $(*)$ as stated uses strict $<$, and at $x = -2, y = -5$ (both negative, hence same sign), $|x + y| = 7 = |x| + |y|$, so $7 < 7$ is false — a fully valid single counterexample to a universal claim. No further pairs need be checked to reject $(*)$.

> **Investigate further:** For which pairs (x, y) does $|x + y| = |x| + |y|$ hold with equality?

8. A student "verifies" "four equal sides $\iff$ square" by checking only that every square has four equal sides.

Answer: B

Method: This checks only necessity (square $\implies$ four equal sides). Sufficiency (four equal sides $\implies$ square) is false: a non-square rhombus has four equal sides but is not a square.

> **Investigate further:** What additional condition, combined with "four equal sides", would give a genuine necessary and sufficient characterisation of a square?

Section D

1. A flawed attempt to prove: if $s^2 \geq 4p$ then there exist real u, v with $u + v = s, uv = p$. *(after TMUA 2021 Paper 2 Q11)*

Answer: B

Method: Every algebraic step (II, III) is correct, but the logical direction is backward. The student assumed u, v exist and derived $s^2 \geq 4p$ from that assumption — this proves the CONVERSE of the theorem, not the theorem itself. A correct proof constructs u, v directly: let u, v be the two roots of $t^2 - st + p = 0$; since the discriminant $s^2 - 4p \geq 0$, these roots are real, and by Vieta's formulas $u + v = s, uv = p$.

> **Investigate further:** Why does the discriminant condition $s^2 - 4p \geq 0$ guarantee the roots of $t^2 - st + p = 0$ are real?

2. A student claims "$n^2 - n + 11$ prime $\implies$ $n^2 + n + 11$ prime" for all positive n, verified for $n = 1, \dots, 9$.

Answer: (a) Finite verification is not a universal proof. (b) $n = 10$.

Method: (a) Checking 9 cases gives no guarantee about $n = 10$ or beyond — a single unchecked value could be a counterexample.
(b) At $n = 10$: $n^2 - n + 11 = 100 - 10 + 11 = 101$, which is prime (premise true). But $n^2 + n + 11 = 100 + 10 + 11 = 121 = 11^2$, which is NOT prime (conclusion false). So the implication fails at $n = 10$, the smallest such value since $n = 1, \dots, 9$ were verified to hold.

> **Investigate further:** Verify that $n^2 - n + 11$ is prime for $n = 1, \dots, 9$ by computing the first few values.

3. Evaluate a contradiction-style proof that "ab even $\implies$ a even or b even".

Answer: D) I and II only

Method: The algebra $(2j + 1)(2k + 1) = 4jk + 2j + 2k + 1 = 2(2jk + j + k) + 1$ is correct, so III is false. Statement I is true: assuming the negation (a, b both odd) and deriving a contradiction (ab odd, contradicting ab even) is a valid contradiction proof. Statement II is also true: the derivation "a, b both odd $\implies$ ab odd" is exactly the contrapositive of the original claim (contrapositive of ab even $\implies$ (a even or b even) is $\neg$(a even or b even) $\implies$ $\neg$(ab even), i.e. "a, b both odd $\implies$ ab odd"), proved here directly without needing the "suppose for contradiction" framing at all.

> **Investigate further:** Rewrite this proof in pure contrapositive form, removing the contradiction framing entirely.

4. Evaluate the "every odd prime > 3 has the form $6k \pm 1$" argument.

Answer: B

Method: Every line is correct: $6k, 6k+2, 6k+4$ are even (Line 2, correctly dismissed); $6k+3 = 3(2k+1)$ is a multiple of 3 exceeding 3 whenever $k \geq 1$, and the case $k = 0$ giving exactly 3 is excluded by the claim's own hypothesis "greater than 3" (Line 3, correctly handled). The proof only ever claimed to show that primes > 3 MUST have the form $6k \pm 1$ (necessity) — it never claimed every number of that form IS prime (sufficiency), and the original claim, read carefully, only asserts the former. So there is no actual gap.

> **Investigate further:** Give an example of a number of the form $6k+1$ that is not prime, to confirm sufficiency genuinely fails.

5. Evaluate: "For all $n \geq 2$, $n^2 - 1$ is composite", proved via $n^2 - 1 = (n-1)(n+1)$.

Answer: (a) Line 3. (b) $n = 2$; and $n = 2$ is the unique exception.

Method: (a) Line 3 is the gap: a product of two integers each "at least 1" is not necessarily composite — compositeness requires BOTH factors to be at least 2. Line 2 only guarantees $n - 1 \geq 1$, not $n - 1 \geq 2$. (b) At $n = 2$: $n^2 - 1 = 3 = 1 \times 3$. The factor of 1 does not make 3 composite — 3 is prime, so the conclusion genuinely fails here. *Proof that $n = 2$ is the only failure*: for every $n \geq 3$, $n - 1 \geq 2$ and $n + 1 \geq 4$, so $n^2 - 1 = (n-1)(n+1)$ is a product of two integers both at least 2, which is precisely the definition of composite — no case analysis or further checking is needed once $n \geq 3$, since the factorisation itself, together with $n - 1 \geq 2$, is a fully general and complete argument covering every such n simultaneously.

> **Investigate further:** Why does the same style of argument fail to show $n^2 - 4$ is composite for all $n \geq 3$? (Check $n = 3$: $n^2 - 4 = 5$.)

Top 5 Patterns Today

1. **The Converse Error (Affirming the Consequent).** Arguing that $P \Rightarrow Q$ and observing Q implies P is the classic converse fallacy; observing a necessary condition holding never proves the sufficient premise. *(see B6, C2, C6, C8, D1)*

2. **The Inverse Error (Denying the Antecedent).** Arguing that $P \Rightarrow Q$ and concluding $\neg P \Rightarrow \neg Q$ confuses an implication with its inverse; multiple distinct conditions often produce the exact same consequence. *(see B8, D3, C2)*

3. **Extraneous Roots from Non-Reversible Operations.** Squaring both sides of $\sqrt{f(x)} = g(x)$ or applying logarithm product rules introduces false solutions where the original expressions were undefined or opposite in sign. *(see A1, A2, A5, B3, B10)*

4. **Division by Zero and Root Loss.** Dividing both sides of an equation by a variable expression (e.g. $x^2 = 3x \Rightarrow x = 3$ or dividing by $a - b$) implicitly assumes that factor is nonzero, erasing valid roots or fabricating false contradictions. *(see A2, A3, B2, D5)*

5. **Incomplete Case Analysis and Boundary Gaps.** Proving a claim only for even n, or ignoring boundary values where factors equal 1 ($n = 2$ in $(n-1)(n+1)$), leaves unexamined cases that often harbour the unique counterexample. *(see B5, B7, C3, D2, D5)*

Common Traps to Avoid

1. **The Fallacy Fallacy (Refuting Argument vs Refuting Claim).** Demonstrating that a proof contains an invalid step shows only that the proof fails, not that the conclusion itself is

false; a true proposition can easily be given an invalid proof. *(see B4, C1, D2)*

2. **Losing Valid Roots via Direct Variable Division.** Cancelling a factor x from $x^2 = kx$ discards the root $x = 0$; always factorise into $x(x - k) = 0$ rather than dividing through by unknown quantities. *(see A2, A3, B2, D5)*

3. **Accepting Extraneous Solutions After Squaring or Logs.** Squaring an equation maps both positive and negative values to positive numbers; all solutions found after squaring or logarithm products must be verified against original domains. *(see A1, A5, B3, B10)*

4. **Assuming "Checking Cases" Equals Proof for Universals.** Verifying that an algebraic property holds for several small integers ($n = 1, 2, 3$) is merely inductive evidence; universal claims fail if even a single case or parity class is omitted. *(see B5, B7, C3, D2)*

5. **Circular Reasoning (Begging the Question).** Assuming the conclusion in order to simplify an intermediate expression produces an argument that appears valid on the surface but is logically vacuous. *(see B9, C4)*

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

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